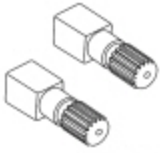
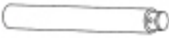


Traction-Lok Clutch Packs

 [Ford Ecat Electronic Parts Catalog](#)

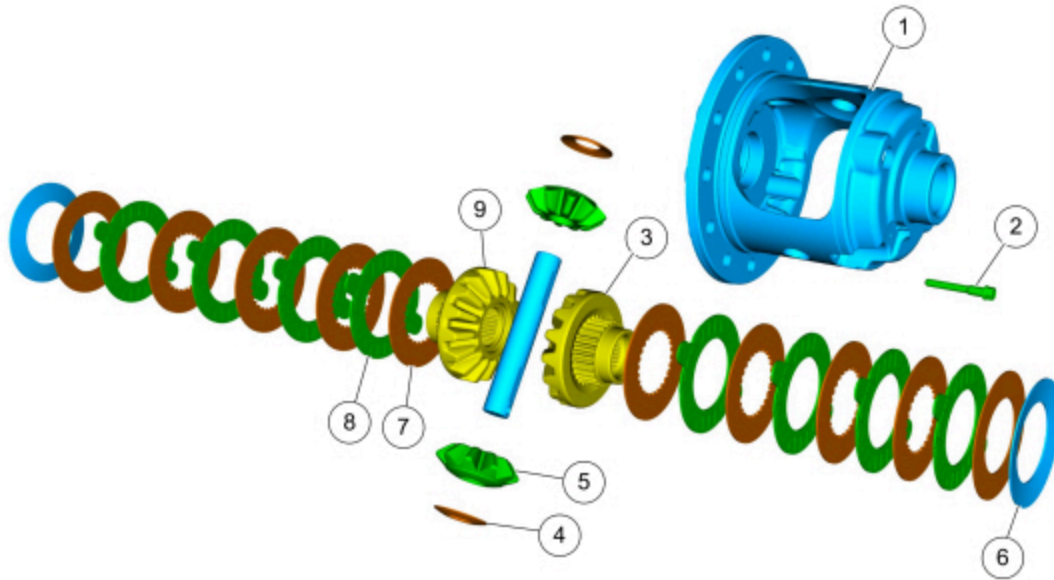
Special Tool(s)

 ST1857-A	Gauge, Differential (Traction-Lock) 205-384 (T97T-4205-A)
 ST1749-A	Gauge, Differential (Traction Lock) 205-386 (T97T-4205-D)
 ST1858-A	Rotator, Differential 205-378 (T97T-4205-C)

Material

Item	Specification
Motorcraft® Additive Friction Modifier XL-3 (US); CXL-3 (Canada)	EST-M2C118-A

Traction-Lok Differential Assembly



N0164250

Item	Part Number	Description
1	4205 4205	Differential case
2	4241 4241	Differential pinion shaft lock bolt (part of 4947)
3	4236 4236	Differential side gear (part of 4215)
4	4230 4230	Differential pinion thrust washer (part of 4215)
5	—	Differential pinion gear (part of 4215)
6	—	Belleville washer
7	4767 4767	Reaction plate (part of 4947)
8	4767 4767	Clutch plate (part of 4947)
9	4211 4211	Differential pinion shaft

Removal

- NOTICE:** New clutch stack assemblies from the kit are pre-measured with the correct shim(s). Do not substitute or install new or different shims to the clutch stack assemblies. Shim selection and measuring is not required.

NOTE: If not replacing the clutch pack, do not mix the differential clutch packs or selective shims from one side with the other.

NOTE: Selective shims are now incorporated as part of the pre-measured Clutch Pack Replacement Kit. Shim selection and measuring is not required.

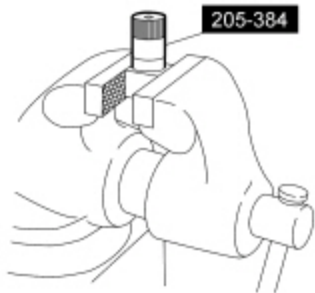
NOTE: The Belleville washer is a dished plate.

If new clutch packs are to be installed, without changing order as packaged soak the clutch stack in clean friction modifier for a minimum of 15 minutes.



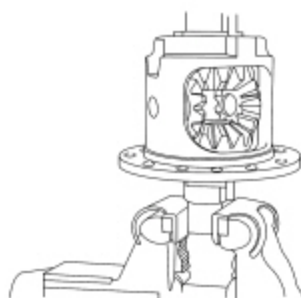
DE0330-B

2. Remove the differential carrier. Tag the differential shims and bearing cups for location. Refer to Differential Carrier.
3. Install the Differential Gauge in a suitable vise.



N0029646

4. Install the differential case on the Differential Gauge.



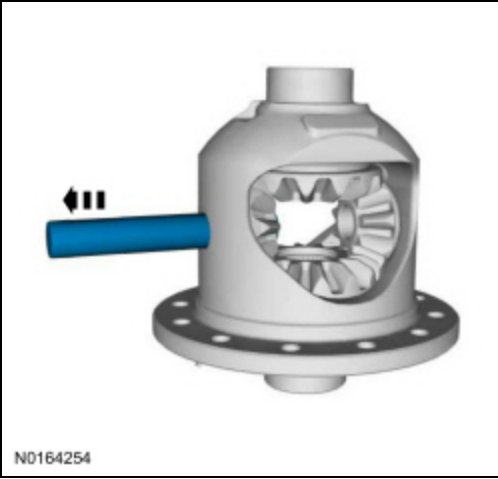
DE0326-B

5. Remove the differential pinion shaft lock bolt.



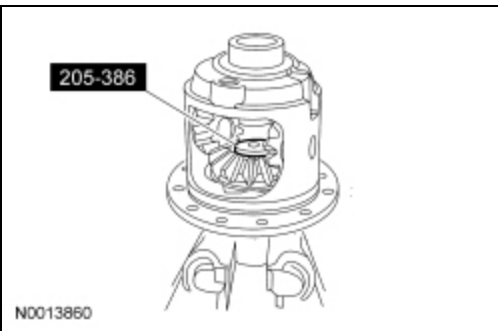
N0164253

6. Remove the differential pinion shaft.

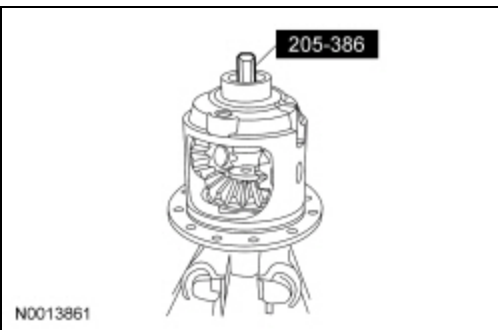


7. **NOTE:** *Apply a small amount of grease to the centering hole of the Differential Gauge.*

Install the Differential Gauge in the bottom differential side gear.

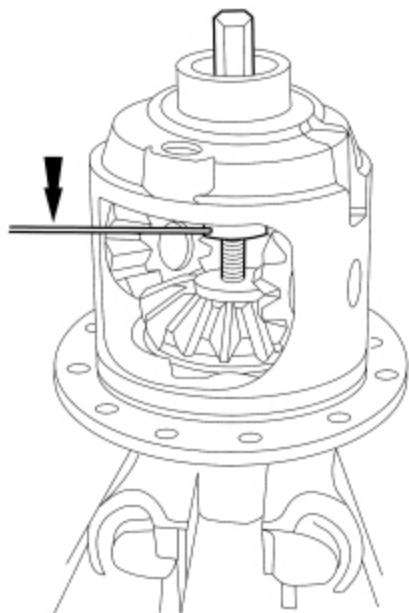


8. Install the nut in the upper differential side gear. Hold the nut in position while installing the hex screw. Tighten the hex-head screw until contact is made with the Step Plate.



9. **NOTE:** *The dowel bar is used to keep the nut from turning when the forcing screw is tightened.*

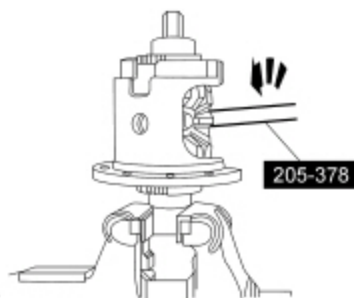
Insert a suitable dowel bar in the hole of the nut. Tighten the forcing screw to force the differential side gears away from the differential pinion gears.



N0013862

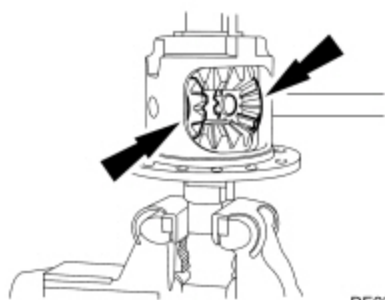
10. **NOTE:** *Differential pinion gear thrust washers cannot be removed independently of the differential pinion gears.*

Insert the Differential Rotator in the pinion shaft bore, and turn the differential case to "walk" the differential pinion gears and differential pinion gear thrust washers out to the differential case windows.



N0029643

11. Remove the differential pinion gears and differential pinion gear thrust washers.



DE0328-A

12. Remove the differential side gears and differential clutch packs with the selective shim and Belleville washer, tag them RH and LH.



DE0329-A

13. **NOTE:** When separating the clutch plates and clutch discs, note the sequence in which they are disassembled. If not replacing the clutches, they must be reassembled in the same sequence as removed.

NOTE: Do not use acids or solvents when cleaning the differential clutch pack. Wipe components with a clean, lint-free cloth only.

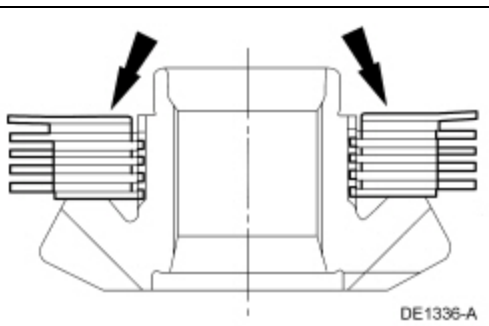
Separate the differential clutch discs and clutch plates for cleaning and inspection.



N0164251

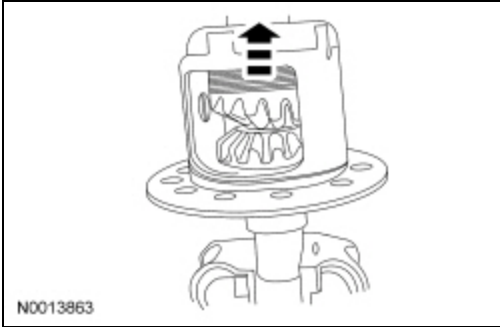
Installation

1. Place the pre-soaked clutch pack and Belleville washer on the differential side gear.
 - The dished or concave side of the Belleville washer must face up against the thrust face of the differential case.



DE1336-A

2. Install the differential side gears with the pre-measured differential clutch packs from the new kit and Belleville washer into the differential case.
 - Hold the differential side gear assembly in place to prevent it from falling out of the differential case.

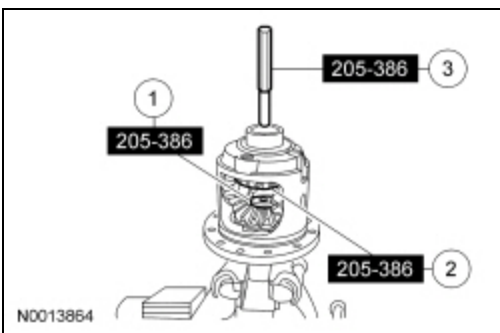


3. **NOTE:** Apply a small amount of grease to the step plate bore.

NOTE: If necessary, insert the dowel bar in the nut bore to keep the nut from turning as the hex screw is tightened.

Assemble the Differential Gauge to the differential case.

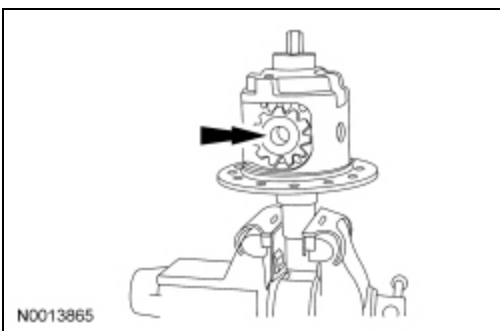
1. Position the Differential Gauge in the bottom differential side gear.
2. Position the Differential Gauge in the top differential side gear and hold it in place.
3. Install the Differential Gauge and tighten it 2 turns after it contacts the bottom step plate.



4. **NOTE:** Lubricate both sides of the differential pinion gear thrust washers with axle lubricant.

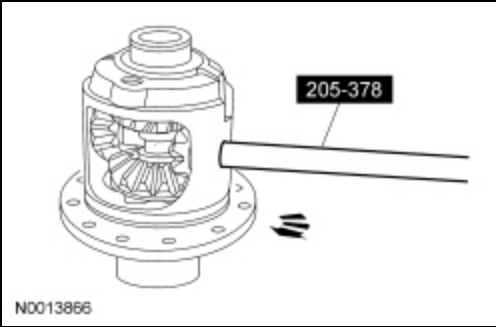
NOTE: Make sure the differential pinion gears are 180 degrees apart so they will align correctly with the pinion shaft bore.

Position the differential pinion gears and differential pinion gear thrust washers in the window of the differential case so they mesh with the differential side gear teeth.



5. **NOTE:** It will probably be necessary to loosen or tighten the forcing screw to allow the differential pinion gears and differential side gears to rotate.

Insert the Differential Rotator into the pinion shaft bore, and turn the differential case. This will cause the differential pinion gears to engage the differential side gears and "walk" into the differential case. Rotate the differential case until the pinion shaft mating holes are lined up exactly with the holes in the differential pinion gears.



6. Remove the Differential Gauge.

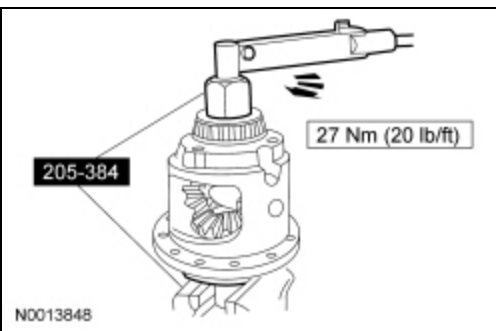
7. **NOTE:** *The differential pinion shaft retaining bolt should only be finger tight at this time.*

Install the differential pinion shaft and the differential pinion shaft lock bolt.



8. Check the torque required to rotate one differential side gear.

- Install the Differential Gauge with the 1/2-inch drive hole as shown.
- The initial break-away torque, if original clutch plates are used, must be at least the minimum specification. The rotating torque required to keep the differential side gear turning with new clutch plates may vary.
- Tighten the differential pinion shaft lock bolt to 34 Nm (25 lb ft).



9. Install the differential carrier. Refer to Differential Carrier.